JJJ-V-002174A(Army-GL)
June 27, 1985
SUPERSEDING
JJJ-V-002174(Army-GL)
June 8, 1983

INTERIM FEDERAL SPECIFICATION

CORN, WHOLE KERNEL, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers whole kernel corn thermostabilized in tray pack cans for use by the Department of Defense as a component of operation rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495 - Coatings, Exterior, for Tinned Food Cans
PPP-B-636 - Boxes, Shipping, Fiberboard

FSC 8915

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Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence Items; General Specifications for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Canned Foods for the Armed Forces

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and human Services

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Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR parts 1-199)

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and Certain Other Processed Food Products

United States Standards for Grades of Canned Whole Kernel (or Whole Grain) Corn

United States Standard for Grades of Frozen Whole Kernel (or Whole Grain) Corn

(Applications for copies should be addressed to the Chief, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

United States Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Market Research and Development Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Environmental Protection Agency (EPA)

National Interim Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water WN550, Environmental Protection Agency, Washington, DC 20460.)

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

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3.2.1 Corn. Corn shall be fresh or frozen in accordance with 3.2.1.1 and 3.2.1.2.

3.2.1.1 Corn, fresh. Fresh corn shall be yellow in color and of a single variety suitable for canning. The corn shall be of such a quality to meet end item grade requirements.

3.2.1.2 Corn, frozen. Frozen corn shall be U.S. Grade A in accordance with the United States for Grades of Frozen Whole Kernel (or Whole Grain) Corn and shall be of the latest season's crop.

3.2.2 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anti-caking agents.

3.2.3 Sugar, granulated. Sugar shall be white refined, granulated cane or beet sugar or a combination thereof and shall comply with MIL-STD-900.

3.2.4 Water. Water is used in the formulation shall conform to the National Interim Primary Drinking Water Regulations.

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Fresh corn preparation. The fresh corn shall be blanched.

3.3.2 Frozen corn preparation. The frozen corn shall be thawed or tempered to the degree necessary for proper filling of the tray pack cans.

3.3.3 Brine preparation. The brine shall be prepared as follows:

a. Twelve pounds of salt and 30 pounds of sugar shall be placed in a proper container and water added until 100 gallons of brine is made.

b. Brine shall be made fresh daily and brine remaining at the end of a production day shall be discarded.

3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with corn and brine such as to conform to the finished product requirements and to the following requirements:

a. Each tray pack can shall be filled with sufficient corn to meet the drained weight requirements of 3.6.

b. Brine may be filled hot or cold.

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c. The cans shall be hermetically sealed with a minimum of 10 inches of mercury (Hg) immediately after filing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 9.6 has been achieved.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not restricted to glass paint, metal, dirt, insects, insect parts, hair, wood, or paper.
b. There shall be no foreign odors or flavors such as but not restricted to rancid, scorched, burnt, stale, sour or moldy.
c. There shall be no color foreign to the product.
d. The average net weight shall be not less than 106 ounces.
e. No individual can shall have a net weight of less than 104 ounces.
f. No individual can shall contain less than 70 ounces of drained corn.
g. The finished product shall be U.S. Grade A in accordance with the U.S. Standards for Grades of Canned Whole Kernel (or Whole Grain) Corn.

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods," and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act, and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

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4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of supervision necessary to assure compliance with the requirements of this specification.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or other applicable purchase document.

4.4.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be evident by examination of pertinent labels, markings, U.S. Grade certificates, certificates of analyses, or other such supplier-furnished valid documents acceptable to the inspection agency. In addition, prior to use, each ingredient shall be examined organoleptically, as necessary, to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, can interior coating, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

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4.6.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be opened and inspected for the defects listed in table II.

TABLE I. Double sampling plan for product inspection. 1/

Lot size (cans)	Sample size (No. of cans)	Cumulative sample	Acceptance number	Rejection number
0-3200	8		0	2
	8	16	1	2
3201-35000	13		0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be acceptable.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product inspection 1/ 2/

Category	Defect
Major	
101	Not grade specified. 3/
102	Drained weight of corn in a can less than 70 ounces.

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair,

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wood, metal, etc), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot. (To be determined by a USDA employee.)

2/ Product not equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an AMS agent.)

3/ The product shall be U.S. Grade A in accordance with the U.S. Standards for Grades of Canned Whole Kernel (or Whole Grain) Corn and inspected in accordance with the provisions of the Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and certain other Processed Food Products.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) Reinforcement with nonmetallic strapping or tape is not used; (4) Height of box liner less than inside depth dimension of box by more than 1/8 inch or extends above top edge of box; (5) Length or width of pads less than box dimensions by more than 1/8 inch; (6) Arrangement or number of cans not as specified.

4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

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5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated with material suitable for the product. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements.

5.1.2 Level C. The product shall be packaged as specified in 5.1.1, except that the cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, and closed in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of grade V3c fiberboard. Each fiberboard box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch).

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The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimension of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined in accordance with 4.7 after submerging the can in boiling water for 30 minutes. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 10 to 15 minutes. Avoid overheating to the point where internal pressure is evident (can shows evidence of bulging). Remove from water and open lid for serving. Caution: Use care when opening as pressure may have been generated within the can.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 15 minutes until contents reach an internal temperature of 165°F.

WARNING: Do not place closed can in oven, as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 25 servings of 1/2 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

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5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3, and 6.3).
(c) Provisions for approved preproduction samples (see 3.6.1 and 6.3).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.509. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection, selection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8915-A787

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
JJJ-V-002174A(ARMY-GL)	CORN, WHOLE KERNEL, THERMOSTABILIZED, TRAY PACK

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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